PG No. 46 $T(c)$ 23 [cubic] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(1,1)}[q](A)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3} + \frac{\sqrt{3}Q_y y}{3} + \frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{Q}}_{1,1}^{(1,-1)}[q](T)$, $\vec{\mathbb{Q}}_{1,2}^{(1,-1)}[q](T)$, $\vec{\mathbb{Q}}_{1,3}^{(1,-1)}[q](T)$

** symmetry

x

y

z

** expression

Q_x

Q_y

Q_z

$\vec{\mathbb{Q}}_{1,1}^{(1,1)}[q](T)$, $\vec{\mathbb{Q}}_{1,2}^{(1,1)}[q](T)$, $\vec{\mathbb{Q}}_{1,3}^{(1,1)}[q](T)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{10}Q_x\left(2x^2 - y^2 - z^2\right)}{10} + \frac{3\sqrt{10}Q_y xy}{10} + \frac{3\sqrt{10}Q_z xz}{10}$$

$$\frac{3\sqrt{10}Q_x xy}{10} - \frac{\sqrt{10}Q_y\left(x^2 - 2y^2 + z^2\right)}{10} + \frac{3\sqrt{10}Q_z yz}{10}$$

$$\frac{3\sqrt{10}Q_x xz}{10} + \frac{3\sqrt{10}Q_y yz}{10} - \frac{\sqrt{10}Q_z\left(x^2 + y^2 - 2z^2\right)}{10}$$

$\vec{\mathbb{G}}_{1,1}^{(1,0)}[q](T)$, $\vec{\mathbb{G}}_{1,2}^{(1,0)}[q](T)$, $\vec{\mathbb{G}}_{1,3}^{(1,0)}[q](T)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{2}Q_y z}{2} - \frac{\sqrt{2}Q_z y}{2}$$

$$-\frac{\sqrt{2}Q_x z}{2} + \frac{\sqrt{2}Q_z x}{2}$$

$$\frac{\sqrt{2}Q_x y}{2} - \frac{\sqrt{2}Q_y x}{2}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E^{(a)})$

** symmetry

$$-\frac{\sqrt{2}x^2}{4} + \frac{\sqrt{6}ix^2}{4} - \frac{\sqrt{2}y^2}{4} - \frac{\sqrt{6}iy^2}{4} + \frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{Q_x x\left(\sqrt{3}-3i\right)}{6}-\frac{Q_y y\left(\sqrt{3}+3i\right)}{6}+\frac{\sqrt{3}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E^{(a)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{iQ_x x\left(3\sqrt{14}x^2+\sqrt{42}ix^2-7\sqrt{14}y^2+\sqrt{42}iy^2-2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}$$
$$+\frac{iQ_y y\left(7\sqrt{14}x^2+\sqrt{42}ix^2-3\sqrt{14}y^2+\sqrt{42}iy^2+2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}+\frac{iQ_z z\left(5\sqrt{14}x^2+3\sqrt{42}ix^2-5\sqrt{14}y^2+3\sqrt{42}iy^2-2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E^{(a)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{Q_x yz\left(\sqrt{3}+i\right)}{2}-\frac{Q_y xz\left(\sqrt{3}-i\right)}{2}-iQ_z xy$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{Q_x x\left(\sqrt{3}+3i\right)}{6}-\frac{Q_y y\left(\sqrt{3}-3i\right)}{6}+\frac{\sqrt{3}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{iQ_x x\left(3\sqrt{14}x^2-\sqrt{42}ix^2-7\sqrt{14}y^2-\sqrt{42}iy^2-2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}$$
$$-\frac{iQ_y y\left(7\sqrt{14}x^2-\sqrt{42}ix^2-3\sqrt{14}y^2-\sqrt{42}iy^2+2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}-\frac{iQ_z z\left(5\sqrt{14}x^2-3\sqrt{42}ix^2-5\sqrt{14}y^2-3\sqrt{42}iy^2+2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{Q_x yz\left(\sqrt{3}-i\right)}{2}-\frac{Q_y xz\left(\sqrt{3}+i\right)}{2}+iQ_z xy$$

$\vec{\mathbb{Q}}_{2,1}^{(1,-1)}[q](T)$, $\vec{\mathbb{Q}}_{2,2}^{(1,-1)}[q](T)$, $\vec{\mathbb{Q}}_{2,3}^{(1,-1)}[q](T)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_y z}{2}+\frac{\sqrt{2}Q_z y}{2}$$

$$\frac{\sqrt{2}Q_x z}{2}+\frac{\sqrt{2}Q_z x}{2}$$

$$\frac{\sqrt{2}Q_x y}{2} + \frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,1)}[q](T)$, $\vec{\mathbb{Q}}_{2,2}^{(1,1)}[q](T)$, $\vec{\mathbb{Q}}_{2,3}^{(1,1)}[q](T)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{7}Q_x xyz}{7} - \frac{\sqrt{7}Q_y z\left(x^2 - 4y^2 + z^2\right)}{7} - \frac{\sqrt{7}Q_z y\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$\frac{\sqrt{7}Q_x z\left(4x^2 - y^2 - z^2\right)}{7} + \frac{5\sqrt{7}Q_y xyz}{7} - \frac{\sqrt{7}Q_z x\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$\frac{\sqrt{7}Q_x y\left(4x^2 - y^2 - z^2\right)}{7} - \frac{\sqrt{7}Q_y x\left(x^2 - 4y^2 + z^2\right)}{7} + \frac{5\sqrt{7}Q_z xyz}{7}$$

$\vec{\mathbb{G}}_{2,1}^{(1,0)}[q](T)$, $\vec{\mathbb{G}}_{2,2}^{(1,0)}[q](T)$, $\vec{\mathbb{G}}_{2,3}^{(1,0)}[q](T)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_x\left(y - z\right)\left(y + z\right)}{2} - \frac{\sqrt{2}Q_y xy}{2} + \frac{\sqrt{2}Q_z xz}{2}$$

$$\frac{\sqrt{2}Q_x xy}{2} - \frac{\sqrt{2}Q_y\left(x - z\right)\left(x + z\right)}{2} - \frac{\sqrt{2}Q_z yz}{2}$$

$$-\frac{\sqrt{2}Q_x xz}{2} + \frac{\sqrt{2}Q_y yz}{2} + \frac{\sqrt{2}Q_z\left(x - y\right)\left(x + y\right)}{2}$$

* Harmonics for rank 3

$\vec{\mathbb{Q}}_{3}^{(1,-1)}[q](A)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_x yz + Q_y xz + Q_z xy$$

$\vec{\mathbb{Q}}_{3}^{(1,1)}[q](A)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_x yz\left(6x^2 - y^2 - z^2\right)}{6} - \frac{\sqrt{15}Q_y xz\left(x^2 - 6y^2 + z^2\right)}{6} - \frac{\sqrt{15}Q_z xy\left(x^2 + y^2 - 6z^2\right)}{6}$$

$\vec{\mathbb{G}}_{3}^{(1,0)}[q](A)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_x x\left(y - z\right)\left(y + z\right)}{2} - \frac{\sqrt{5}Q_y y\left(x - z\right)\left(x + z\right)}{2} + \frac{\sqrt{5}Q_z z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](T,1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](T,1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,-1)}[q](T,1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{15}Q_x\left(2x^2 - y^2 - z^2\right)}{10} - \frac{\sqrt{15}Q_yxy}{5} - \frac{\sqrt{15}Q_zxz}{5}$$

$$-\frac{\sqrt{15}Q_xxy}{5} - \frac{\sqrt{15}Q_y\left(x^2 - 2y^2 + z^2\right)}{10} - \frac{\sqrt{15}Q_zyz}{5}$$

$$-\frac{\sqrt{15}Q_xxz}{5} - \frac{\sqrt{15}Q_yyz}{5} - \frac{\sqrt{15}Q_z\left(x^2 + y^2 - 2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](T,2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](T,2)$, $\vec{\mathbb{Q}}_{3,3}^{(1,-1)}[q](T,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{Q_x\left(y - z\right)\left(y + z\right)}{2} + Q_yxy - Q_zxz$$

$$-Q_xxy - \frac{Q_y\left(x - z\right)\left(x + z\right)}{2} + Q_zyz$$

$$Q_xxz - Q_yyz + \frac{Q_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,1)}[q](T,1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,1)}[q](T,1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,1)}[q](T,1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{Q_x\left(8x^4 - 24x^2y^2 - 24x^2z^2 + 3y^4 + 6y^2z^2 + 3z^4\right)}{12} + \frac{5Q_yxy\left(4x^2 - 3y^2 - 3z^2\right)}{12} + \frac{5Q_zxz\left(4x^2 - 3y^2 - 3z^2\right)}{12}$$

$$-\frac{5Q_xxy\left(3x^2 - 4y^2 + 3z^2\right)}{12} + \frac{Q_y\left(3x^4 - 24x^2y^2 + 6x^2z^2 + 8y^4 - 24y^2z^2 + 3z^4\right)}{12} - \frac{5Q_zyz\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$

$$-\frac{5Q_xxz\left(3x^2 + 3y^2 - 4z^2\right)}{12} - \frac{5Q_yyz\left(3x^2 + 3y^2 - 4z^2\right)}{12} + \frac{Q_z\left(3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,1)}[q](T,2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,1)}[q](T,2)$, $\vec{\mathbb{Q}}_{3,3}^{(1,1)}[q](T,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}Q_yxy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}Q_zxz\left(2x^2+9y^2-5z^2\right)}{12}$$

$$-\frac{\sqrt{15}Q_xxy\left(5x^2-2y^2-9z^2\right)}{12}+\frac{\sqrt{15}Q_y\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}Q_zyz\left(9x^2+2y^2-5z^2\right)}{12}$$

$$\frac{\sqrt{15}Q_xxz\left(5x^2-9y^2-2z^2\right)}{12}+\frac{\sqrt{15}Q_yyz\left(9x^2-5y^2+2z^2\right)}{12}-\frac{\sqrt{15}Q_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](T,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](T,1)$, $\vec{\mathbb{G}}_{3,3}^{(1,0)}[q](T,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{3}Q_yz\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}Q_zy\left(4x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{3}Q_xz\left(x^2-4y^2+z^2\right)}{4}-\frac{\sqrt{3}Q_zx\left(x^2-4y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{3}Q_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](T,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](T,2)$, $\vec{\mathbb{G}}_{3,3}^{(1,0)}[q](T,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\sqrt{5}Q_xxyz+\frac{\sqrt{5}Q_yz\left(2x^2+y^2-z^2\right)}{4}+\frac{\sqrt{5}Q_zy\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_xz\left(x^2+2y^2-z^2\right)}{4}-\sqrt{5}Q_yxyz-\frac{\sqrt{5}Q_zx\left(x^2-2y^2-z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_xy\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}Q_yx\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}Q_zxyz$$